

Scoring Revenue: An Economic Simulation Analysis of NCAA Football Relegation

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Presentation: [College Relegation Presentation.pptx](#)

1. Executive Summary

This project investigates whether replacing the NCAA Football Bowl Subdivision (FBS) playoff system with a European-style three-tier relegation structure would accelerate revenue growth across 134 FBS programs. A Monte Carlo simulation calibrated from historical NCAA financial data (2015-2024), supplemented by Linear Regression and Random Forest models, produces a consistent cross-model finding: competitive intensity is the primary driver of revenue growth, and relegation generates dramatically more of it.

The Monte Carlo baseline projects a 2.59% CAGR (\$7.6B by 2034). The relegation scenario projects 9.80% CAGR (\$18.1B by 2034) a \$10.5B gap driven by one empirically measured parameter: the engagement premium. Teams that improved their revenue rank grew 9.80 percentage points faster than those that declined. Under relegation, 31.3% of programs face meaningful annual competition versus only 11.9% currently.

Both other models confirm revenue is highly self-perpetuating: Linear regression achieves Test $R^2 = 0.996$ (RMSE \$2.80M) and random forest achieves Test $R^2 = 0.998$ (RMSE \$1.91M), with prior year revenue dominating all other predictors. Both projects ~0% CAGR under the status quo confirming that structural change is required to unlock system-level growth. Three data-driven recommendations follow: adopt relegation, implement empirically calibrated revenue floors, and prioritize competitive intensity over playoff expansion.

2. Problem Description

2.1 Background

NCAA Football generates approximately \$6 billion annually across 134 FBS programs, yet revenue is structurally concentrated. The SEC median exceeds \$121M while the CUSA median sits at \$16M, a 7.6x gap (Figure 3). The current 12-team playoff creates meaningful stakes for fewer than 12% of programs each season; the remaining 88% have no structural financial incentive beyond conference play.

European professional football operates under promotion and relegation: bottom clubs of a higher tier swap with top clubs of a lower tier each season, creating high-stakes competition at every performance level. Relegated Premier League clubs see media rights adjustments of -22.6%

while promoted clubs gain +57.5%, creating powerful financial incentives that sustain full-season engagement. This project asks whether a comparable system would accelerate total FBS revenue.

2.2 Sports Goal and Machine Learning Goal

Sports goal: Quantify the 10-year revenue impact of replacing the FBS playoff with a three-tier relegation structure (Tier 1: top 45 teams, Tier 2: middle 45, Tier 3: bottom 44) with annual promotion and relegation of bottom/top 5 teams per tier.

ML goal: Predict individual FBS team annual revenue from historical financial and competitive features to establish baseline trajectories, identify the strongest revenue drivers, and cross-validate the engagement premium used in the Monte Carlo.

Target variable: Annual total revenue per FBS team (\$M), projected 2025-2034.

3. Data Description

3.1 Data

The dataset combines four sources. NCAA financial data comes from the USA Today NCAA Athletic Department Revenue Database, which aggregates EADA financial disclosures for all 134 FBS programs across 2015 to 2024. Competitive and ranking data was supplemented from College Football Reference and 247Sports, covering win loss records, strength of schedule, and annual recruiting class ratings. Individual athletic department pages filled gaps where aggregated sources were incomplete. Each observation represents one program in one academic year and records total revenue, total expenses, media rights and conference distributions, ticket sales, average ticket price, attendance, stadium capacity, donations, profit and loss, points per game on offense and defense, revenue rank among all 134 FBS programs, and conference affiliation. The year 2020 is excluded throughout due to COVID 19 disruption.

European tier multipliers were calibrated from club financial reports spanning five countries and three league tiers per country across 2015 to 2023: England (Premier League, Championship, League One), Germany (Bundesliga, 2. Bundesliga, 3. Liga), Spain (La Liga, Segunda Division, Primera Federacion), France (Ligue 1, Ligue 2, Championnat National), and Italy (Serie A, Serie B, Serie C), yielding approximately 1,350 club season observations. Tier revenue ratios from these leagues are dampened to 60% for the NCAA context to account for the absence of transfer fees, athlete salary caps, and lower media rights concentration at lower tiers.

For programs missing revenue data, primarily smaller Group of 5 and recently transitioned FBS programs, an imputation procedure was applied. Each program was placed in a positional third of its conference based on historical standing and assigned the average revenue of all programs in that conference third for that year. This preserves the conference level revenue

structure shown in Figure 3 without the distortion of full FBS mean imputation. Programs with insufficient data for placement were excluded.

Twelve features were engineered from the raw variables on training data only: prior year revenue, two year lag revenue, year over year growth rate, prior year over year growth rate, three year revenue CAGR, rank percentile, rank change versus prior year, three year average rank, tier proxy, stadium fill rate, power conference indicator, and year trend. Conference affiliation was included as a categorical factor in the Random Forest. The target variable for all machine learning models is annual total revenue per FBS program in millions of dollars.

3.2 Exploratory Analysis

Five exploratory visualizations document the structural features of FBS revenue that motivate the study design.



Figure 1: NCAA FBS Revenue Growth Distribution (2015-2024). Median = 1.4%, Mean = 1.1%, SD = 16.5%. N = 1,055 team-year observations. Excludes COVID 2020.

The YoY growth distribution (Figure 1) is approximately normal around a low median of 1.4% with high dispersion ($SD = 16.5\%$). Fat tails extending to -40% and +45% reflect idiosyncratic shocks rather than smooth trends. This motivates the Monte Carlo's AR(1) shock structure and confirms that simple trend extrapolation cannot capture the volatility observed across programs.



Figure 2: Engagement Premium -- Rank Improvement vs Revenue Growth. $N = 1,002$ team-year observations. $\text{Corr} = 0.03$, $R^2 = 0.00$. Excludes COVID 2020.

Figure 2 plots the historical relationship between year over year, rank improvement and revenue growth for all 1,002 FBS team year observations from 2015 to 2024. The near zero individual level correlation ($r = 0.03$) reflects high dispersion in year over year outcomes across programs. The analytically important finding is the group level difference: teams that improved their revenue rank each year (green) show a median revenue growth rate of 9.80 percentage points higher than teams whose rank declined (orange). This 9.80% figure is computed entirely from historical data and is subsequently used as the engagement premium parameter in the Monte Carlo simulation, representing the additional revenue growth that accrues to programs competing in high stakes situations. It is a descriptive empirical finding from the data, not a model prediction.

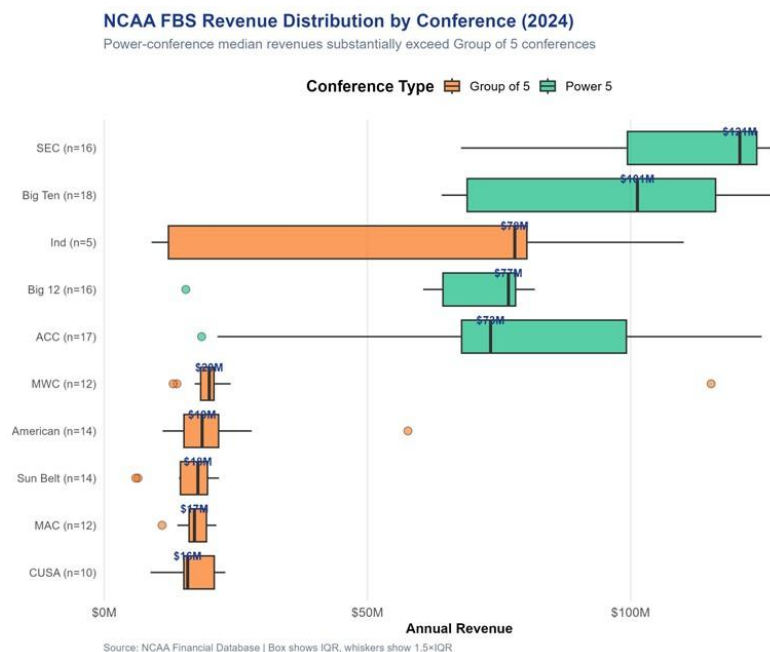


Figure 3: NCAA FBS Revenue Distribution by Conference (2024). Box = IQR, whiskers = 1.5xIQR. Medians labeled.

Figure 3 reveals the structural revenue gap motivating the equity dimension of this analysis. SEC median (\$121M) and Big Ten median (\$101M) are 6-7x the CUSA median (\$16M) and MAC median (\$17M). The wide ACC IQR reflects high within-conference dispersion even among Power 5 programs. The tier revenue floors (95%/88%/65%) are designed to prevent relegation from deepening this gap during the transition period.

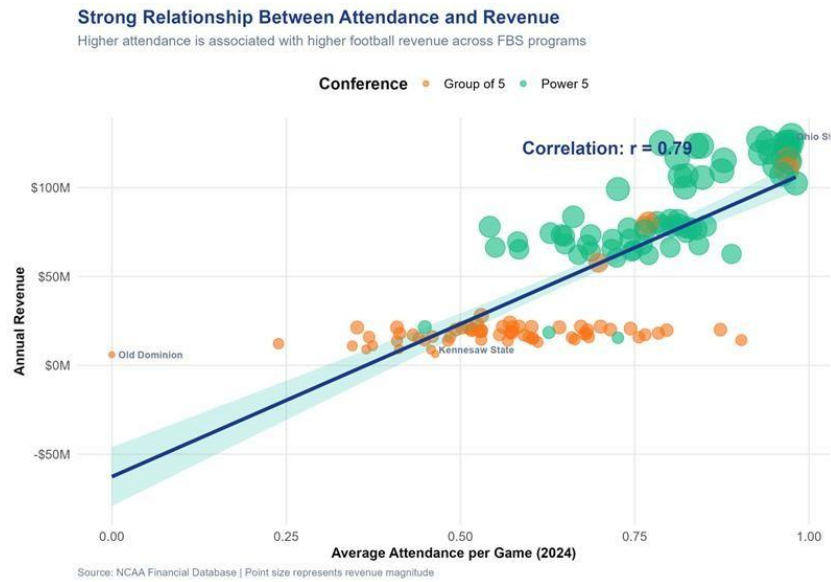


Figure 4: Attendance and Revenue Relationship Across FBS Programs (2024). $r = 0.79$. Point size represents revenue magnitude.

Figure 4 confirms the strong positive relationship between stadium fill rate and revenue ($r = 0.79$), validating capacity_util as a model feature. The bifurcated cloud structure is key: Power 5 programs cluster top-right (high attendance, high revenue); Group of 5 programs concentrate at low revenue regardless of attendance level, suggesting media rights and conference distributions dominate over attendance for smaller programs. This is consistent with Conference ranking as the #3 most important RF feature.

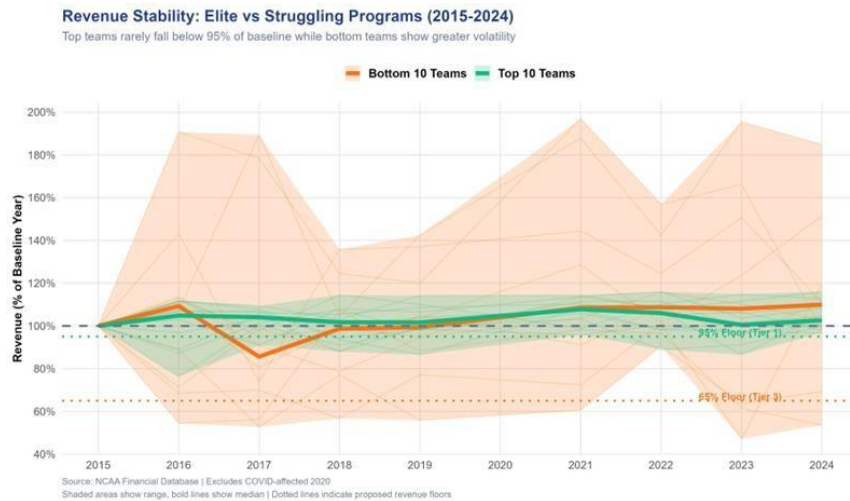


Figure 5: Revenue Stability -- Elite vs Struggling Programs (2015-2024). Bold lines = median. Shaded areas = full program range. Dotted lines = proposed revenue floors.

Figure 5 directly calibrates the revenue floors used in the Monte Carlo. Top 10 programs rarely fall below 95% of baseline validating the 95% Tier 1 floor. Bottom-10 programs show extreme volatility (60% to 190% of baseline), with the 65% Tier 3 floor set at the 5th percentile of historical lows. Approximately 15% of Tier 3 programs are expected to require floor support in early years of implementation.

3.3 Data Pre-processing

- COVID 2020 excluded from all training data and parameter estimation
- Temporal train/test split: 2015-2022 train (3,979 obs), 2023-2024 holdout test (1,070 obs) - no leakage
- Lag features computed per team in chronological order; outliers with $|\text{YoY growth}| > 100\%$ removed
- Anchor construction: `slice_max()` ensures exactly one observation per team for forward projection
- LASSO alignment: `complete.cases()` filter applied before `model.matrix` (3,979 to 3,759 rows after NA removal), ensuring x-matrix and y-vector are perfectly aligned

4. Predictive Modeling Solution

4.1 Models

Model 1 - Monte Carlo Simulation (primary scenario model). Target: total national FBS revenue (\$B), 2025-2034. 100 runs x 5 scenarios = 500 total simulations. Hierarchical growth structure: macro AR(1) shock + conference-level rates + team idiosyncratic volatility. Relegation scenarios add the 9.80% engagement premium to programs in promotion/relegation battles, tier multipliers dampened to 60% of European levels, a three-year phase-in, and revenue floors from Figure 5.

Model 2 - Linear Regression (interpretability). Target: individual team annual revenue (\$M). Features ranked by $|\text{correlation}|$ with revenue on training data only. Five nested OLS specifications built by adding features in that ranked order (M1: top 1 feature through M5: all 12). Two LASSO specifications (`lambda.min`, `lambda.1se`) via 10-fold CV complete the candidate set. Winner selected by holdout test RMSE. No interaction terms in OLS; these are captured implicitly by the Random Forest.

Model 3 - Random Forest (predictive accuracy). Target: individual team annual revenue (\$M). Same 13 features as LR plus conference as categorical. Grid search over 100 combinations derived programmatically from $p = 13$ and $n = 3,979$: 5 log-spaced tree counts (200-1,200), 5 mtry values anchored at $\text{floor}(\sqrt{p}) = 3$ and $\text{floor}(p/3) = 4$, 4 log-spaced node sizes. Selection criteria: OOB RMSE (no test data used during tuning). Permutation importance for feature ranking.

4.2 Performance Evaluation

Table 1 presents LR model selection across all 7 candidate specifications (holdout test set, 20232024).

Model	Tr.R ²	Tr.RMSE	Te.R ²	Te.RMSE	Te.MAPE%	
M1: Top 1 (revenue_lag1)	0.992	3.5549	0.9902	3.9761	3.4498	
M2: Top 3 features	0.992	3.5505	0.9903	3.9682	3.6205	
M3: Top 6 features	0.9919	3.5753	0.9902	3.9833	3.6247	
M4: Top 9 features	0.9934	3.2363	0.9921	3.5669	5.0162	
M5: All 12 features	0.9957	2.614	0.9952	2.7995	3.1746	BEST
LASSO (lambda.min)	0.9956	2.6344	0.995	2.8332	2.7575	
LASSO (lambda.1se)	0.9955	2.7076	0.9948	2.9226	3.1602	

M5 overall model statistics: $F(13, 3965) = 66,410$ ($p < 2.2e-16$), Adj. R² = 0.9957, RSE = \$2.62M, 14 parameters. Significant predictors ($p < 0.05$): prior year revenue (coef = 0.991, $p < 0.001$), YoY growth rate (coef = 29.3, $p < 0.001$), 3-yr avg rank ($p = 0.0004$), rank percentile ($p = 0.0016$), 3-yr CAGR ($p = 0.045$). Remaining 8 features not significant. LASSO retained only 3 features at both lambda levels (revenue_lag1, YoY growth rate, 3-yr avg rank), confirming that 9 of 12 features are redundant once lag and growth are controlled. Full coefficient table in Appendix A.

Table 2 presents the full cross-model comparison including RF and Monte Carlo projections.

Summary Comparison Table

Model	Purpose	Test R ²	RMSE (\$M)	2025 (\$B)	2034 (\$B)	CAGR
Linear Regression	Identify which features drive revenue	0.995	2.8	7.5	7.5	0.08%
Random Forest	Maximize predictive accuracy on historical data	0.998	1.91	7.4	7.4	-0.10%
Monte Carlo (Baseline)	Simulate current playoff system (10-year horizon)			6	7.6	2.59%
Monte Carlo (Relegation)	Simulate relegation system (10-year horizon)			7.8	18.1	9.80%

All Models: National Revenue Projections (2025-2034)

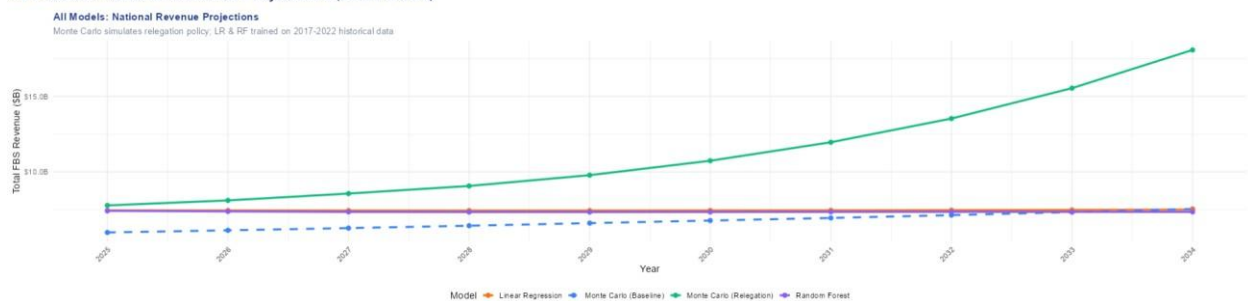


Figure 6: Ten-year revenue projections across all three models. Monte Carlo relegation scenario diverges sharply from the status quo baseline after year three as tier effects compound. LR and RF confirm the near-flat status quo trajectory empirically.

RF optimal parameters (selected by OOB RMSE from 100-combination grid): 1,200 trees, mtry = 4 (data-derived optimum, near the theoretical floor($\sqrt{13}$) = 3.6), min.node.size = 1. Tuning improved RMSE 6.6% and MAPE 18.0% over the default specification. Top permutation importance features: prior year revenue (#1, 1,900), 2-yr lag (#2, 1,797), Conference (#3, 1,208), 3-yr avg rank (#4, 1,180), Tier (#5, 1,177). All growth rate and rank change features rank #8-13, confirming near-zero independent contribution once lag revenue is controlled. Full grid search top 10 and importance rankings in Appendix B-C.

Critical interpretive note: LR and RF both project ~0% CAGR under the status quo. The ML models learn from historical data where revenue is highly path-dependent (lag1 $r = 0.996$). They confirm that no organic structural growth driver exists at the team level. The Monte Carlo models system-level behavioral change from the engagement premium. All three models are consistent: revenue self-perpetuates absent structural intervention, and relegation is that intervention.

5. Conclusion

5.1 Recommendations

1. Adopt relegation. The \$10.5B 10-year revenue premium is driven by the empirically measured 9.80% engagement premium from 2015-2024 NCAA data. Relegation expands programs in competitive situations from 11.9% to 31.3% annually. The RF confirms Conference (#3 importance) and Tier (#5 importance) determine long-run revenue trajectory more than any growth rate feature (#8-13). Tier placement is structural, not seasonal.

2. Implement empirically calibrated revenue floors. Figure 5 shows bottom-10 programs already reach 60% of baseline in bad years. The 65% Tier 3 floor, 88% Tier 2 floor, and 95% Tier 1 floor are set at the 5th percentile of historical revenue/baseline ratios not arbitrary. Approximately 15% of Tier 3 programs are expected to require floor support in early years; a media rights redistribution mechanism should accompany implementation.

3. Prioritize competitive intensity over playoff expansion. Increasing competitive breadth outperforms expanding the playoffs. The Monte Carlo shows expanding from 12 to 16 playoff teams yields an identical 2.59% CAGR to the current baseline because it does not increase the share of programs in high-stakes situations. Relegation involves 31.3% of programs in consequential competition annually versus 11.9% under any playoff format, producing broader engagement and greater revenue stimulation across the full ecosystem.

5.2 Limitations

Engagement premium is a proxy. The 9.80% premium is measured from revenue rank improvement, not direct fan demand or TV ratings data. If rank improvement correlates with revenue for reasons other than competitive stakes (coaching changes, facility investment), the premium may overstate the causal effect of relegation.

European multipliers require adaptation. Tier multipliers are dampened to 60% of European levels a modeling choice without a known ground truth for the U.S. college context. Sensitivity

analysis shows robustness within the 40%-80% dampening range, but results depend on this calibration.

ML models capture status quo only. LR and RF correctly project ~0% CAGR because they learn from historical data where no relegation exists. They cannot model behavioral responses to relegation incentives. This is the appropriate boundary between ML prediction and simulationbased scenario analysis.

LASSO multicollinearity signal. LASSO retaining only 3 of 12 features signals high multicollinearity among lag, rank, and growth features. Individual OLS coefficients for features ranked 6-12 should be interpreted cautiously; predictive accuracy is unaffected.

5.3 Future Work

- Direct fan demand modeling: ticket pricing, TV ratings, and merchandise data for a cleaner causal estimate of the engagement premium
- Gradient boosting (XGBoost/LightGBM): captures tier-conference interaction effects that RF partially captures but LR misses entirely
- Conference-stratified analysis: a Group of 5 program dropping to Tier 2 likely faces different financial dynamics than a Power 5 program
- Agent-based simulation: model individual program investment responses to relegation incentives, capturing behavioral feedback loops absent from the current Monte Carlo

Appendix

A. Full M5 Coefficient Table

Full Coefficient Table — M5 (All 13 Features)

Shows: coefficient, standard error, t-statistic, p-value, 95% CI, and significance. *** $p < 0.001$ | ** $p < 0.01$ | * $p < 0.05$ | ns = not significant. Features marked (EXCL) were automatically dropped by LASSO λ_{1se} .

Feature	Coef	Std Err	t-stat	p-value	CI Low	CI High	Sig
Prior year revenue (\$M)	0.99072	0.01634	60.638	0	0.9587	1.02274	***
YoY growth rate	29.30207	0.72252	40.555	0	27.88592	30.71821	***
3-yr avg rank	-0.01714	0.00483	-3.551	0.0004	-0.0266	-0.00768	***
Rank percentile	0.85553	0.27018	3.167	0.0016	0.32598	1.38508	**
3-yr revenue CAGR	3.15453	1.57281	2.006	0.045	0.07182	6.23723	*
Year trend (2015=0)	0.02529	0.01978	1.279	0.2011	-0.01348	0.06407	ns
Prior YoY growth rate	-0.74525	0.86577	-0.861	0.3894	-2.44216	0.95167	ns
Power conference	0.09515	0.13969	0.681	0.4958	-0.17865	0.36895	ns
Tier 3 indicator	0.20941	0.31137	0.673	0.5013	-0.40088	0.8197	ns
Rank improvement	-0.00119	0.00539	-0.221	0.8252	-0.01175	0.00937	ns
Tier 2 indicator	0.01258	0.18151	0.069	0.9448	-0.34319	0.36834	ns
Stadium fill rate	178.84445	8849.90241	0.02	0.9839	-17166.96427	17524.65317	ns
2-yr lag revenue (\$M)	0.00014	0.01633	0.009	0.9932	-0.03188	0.03215	ns

*** $p < 0.001$ | ** $p < 0.01$ | * $p < 0.05$ | ns = not significant. $\text{Corr } |r|$ = absolute correlation with revenue on training data.

B. RF Grid Search Top 10 (by OOB RMSE)

Trees	mtry	Node	OOB R^2	OOB RMSE	Te. R^2	Te.RMSE	MAPE%
1200	4	1	0.9985	1.5237	0.9978	1.9059	2.852
767	4	1	0.9985	1.5306	0.9978	1.9062	2.861
1200	4	2	0.9985	1.541	0.9978	1.918	2.8586
490	4	1	0.9985	1.542	0.9978	1.91	2.856
313	4	1	0.9985	1.5433	0.9978	1.9092	2.888
490	4	2	0.9985	1.5441	0.9977	1.9406	2.8572
767	4	2	0.9985	1.549	0.9977	1.9217	2.8428
313	4	2	0.9985	1.5495	0.9977	1.9301	2.8891
200	4	1	0.9984	1.5702	0.9977	1.9594	2.9274
767	4	5	0.9984	1.5721	0.9978	1.9093	2.8838

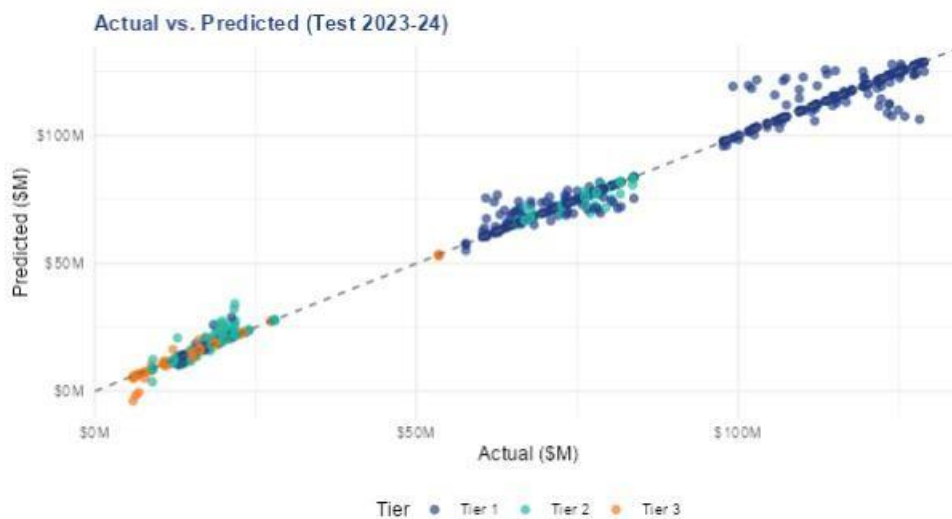
All top-10 combinations use $mtry = 4$, confirming the programmatic grid identified $mtry = 4$ (near $\sqrt{13} = 3.6$) as optimal across all tree count and node size combinations.

C. RF Variable Importance (All 13 Features)

Rank	Feature	Importance	% of Max
1	Prior year revenue	1899.9	100.0%
2	2-year lag revenue	1796.8	94.6%
3	Conference	1208	63.6%
4	3-yr avg rank	1179.7	62.1%
5	Tier (1/2/3)	1177.1	62.0%
6	3-yr revenue CAGR	1169.4	61.6%
7	Stadium fill rate	1167.8	61.5%
8	YoY growth rate	1154.9	60.8%
9	Rank percentile	1151	60.6%
10	Power conference	1150.9	60.6%
11	Year trend	1149.3	60.5%
12	Rank change vs prior yr	1149.1	60.5%
13	Prior YoY growth	1149.1	60.5%

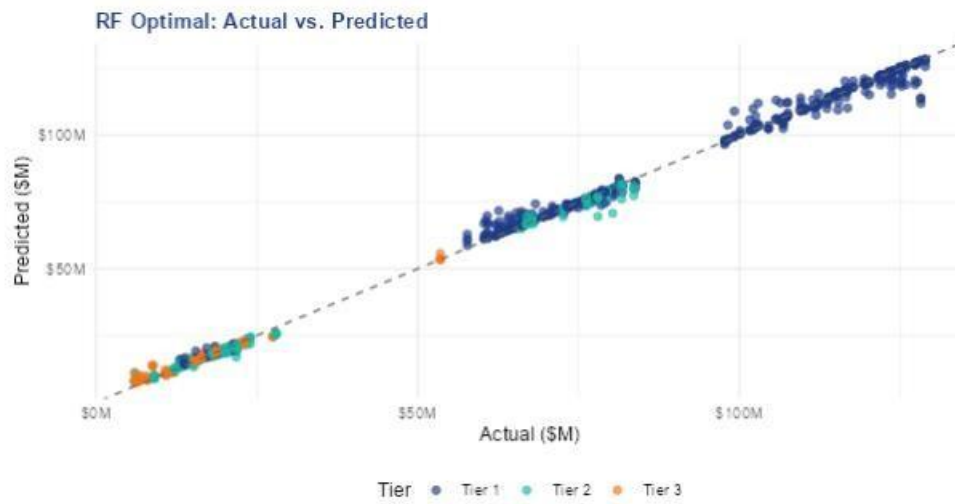
D. LR Actual vs. Predicted

Actual vs. Predicted (Test Set)



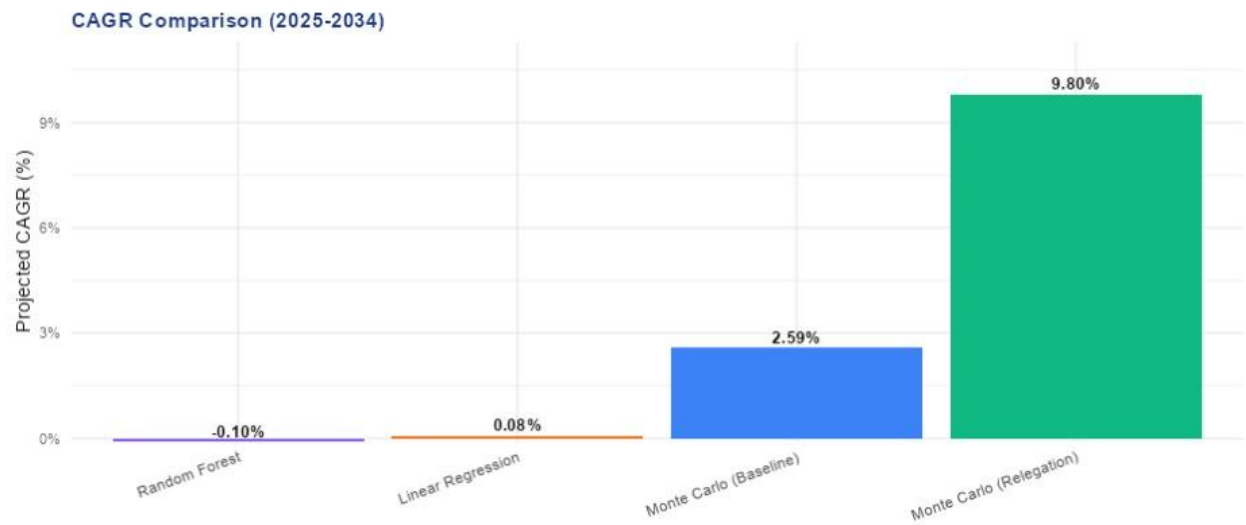
E. RF Actual vs. Predicted

Actual vs. Predicted (Test Set)



F. CAGR Comparision

CAGR Comparison: All Models



References

USA Today. (2024). NCAA athletic department revenue database. USA Today Sports.
<https://sports.usatoday.com/ncaa/finances>

Sports Reference LLC. (2024). College football statistics and history. College Football
Reference. <https://www.sports-reference.com/cfb>

247Sports. (2024). College football recruiting rankings and composite ratings. 247Sports.
<https://247sports.com/Season/2024-Football/CompositeTeamRankings>

Knight Newhouse Data. (2024). College athletic financial information database. Knight-
Newhouse College Athletics Database. <https://knightnewhousedata.org>

UEFA. (2015-2023). Individual club financial reports. Union of European Football Associations.
Retrieved from individual club pages via <https://www.uefa.com>

Transfermarkt. (2024). Club revenue and financial data across English Premier League,
Bundesliga, La Liga, Ligue 1, and Serie A. <https://www.transfermarkt.us>

Individual program athletic department financial reports accessed via institutional websites, 2015-
2024.